

WEEKLY PROGRESS REPORT – WEEK 1

Project Title

Healthcare Analytics Pipeline for MediCare Hospital

Group Number: 3

Team Members

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Week 1 Objectives

The objective of Week 1 was to understand the project requirements, identify suitable healthcare datasets, define the project scope, select technologies, and prepare an implementation plan for the Healthcare Analytics Pipeline.

Activities Completed

1. Project Discussion and Planning

The team conducted multiple discussions to understand the project requirements, expected deliverables, timeline, and implementation approach. The project objectives, scope, architecture, and workflow were finalized during these discussions.

2. Problem Understanding

The team analyzed the healthcare analytics problem and identified the need for an end-to-end data engineering pipeline capable of handling healthcare data ingestion, processing, storage, analytics, machine learning, and visualization.

3. Dataset Identification

Relevant healthcare datasets were explored and evaluated. A healthcare dataset containing patient information, disease details, treatment costs, admission status, and admission dates was selected for the project.

4. Dataset Schema Analysis

The dataset structure was reviewed and documented.

Dataset Attributes:

- Patient ID
- Patient Name
- Age
- Gender

- Disease
- City
- Treatment Cost
- Admission Status
- Admission Date

5. Technology Stack Selection

The team finalized the technology stack required for project implementation:

- Python
- Pandas
- PostgreSQL
- AWS S3
- AWS RDS PostgreSQL
- Apache Airflow
- Apache Spark (PySpark)
- Streamlit
- Scikit-Learn
- Docker
- GitHub

6. High-Level Architecture Design

A preliminary architecture was designed to define the complete project workflow.

Proposed Workflow:

Healthcare Dataset → AWS S3 → ETL Pipeline → Apache Airflow → PySpark → AWS RDS PostgreSQL → Data Warehouse → Machine Learning → Streamlit Dashboard

7. Team Coordination and Responsibility Allocation

The team discussed the project execution strategy and distributed responsibilities among members. All members participated in planning, documentation, validation, testing, review activities, and project discussions throughout the week.

Deliverables Completed

- Project requirement analysis
- Dataset selection
- Dataset schema overview
- Technology stack finalization

- Initial architecture design
- Project implementation plan
- Team responsibility discussion

Challenges Faced

- Understanding healthcare analytics requirements
- Evaluating suitable datasets
- Selecting appropriate cloud and analytics technologies
- Designing a scalable architecture

Plan for Week 2

- Perform data cleaning and preprocessing
- Handle missing values and validation checks
- Standardize dataset formats
- Prepare data dictionary
- Begin ETL pipeline development

Conclusion

Week 1 focused on project planning, dataset analysis, architecture design, and technology selection. The team successfully established the foundation required for implementing the Healthcare Analytics Pipeline and prepared a structured roadmap for subsequent phases.